PRE-DEFENSE 1 · 24 OCTOBER 2026

Development of an Intelligent Order Management System with Digital Twin Integration for Furniture Manufacturing

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PROJECT PROPOSAL

An unreliable delivery date starts at the point of sale

Every order is a small engineering task

Dimensions, materials, finishes and fittings vary, and showroom data must become production-ready data.

The date is promised too early

The customer expects a date before anyone has checked factory capacity or material availability.

The cost comes later

Missed deadlines, rescheduling and expediting. A better commitment is needed at the moment of sale.

TARGET USERS

Showroom managers, who give the date to the customer, and production planners, who see the factory load.

PROJECT PROPOSAL

Aim, objectives and scope

> Design and evaluate an order-management system with a simulation-based digital twin that improves the accuracy of delivery-date commitments at order acceptance in customized furniture manufacturing.

Objectives

- **O1** Define the digital order
- **O2** Design architecture and data
- **O3** Build a calibrated twin
- **O4** Implement API, twin and web UI
- **O5** Compare with baselines B1, B2

Hypotheses

H1 The mean absolute error of the committed date is smaller than for B1 and for B2.

H2 A larger share of orders is finished by the committed date.

Out of scope

- Product configurator and 3D view
- Pricing and engineering design
- Shop-floor schedule optimisation
- Connection to real machines
- Deployment in a live factory

EVIDENCE OF RELEVANCE

Practitioners at LignaTech describe the same problem

WHO WE ASKED

LignaTech

Workshop and salon making custom doors and furniture

Production owner
Salon manager

About 20 years in the field.
Anonymous, with consent: no names recorded.

Date from many sources

The manager checks with production. Load and materials sit in tables, messages and calls.

Delays have known causes

Late materials, workshop load, breakdowns, changes after approval, measuring errors.

A missed date is costly

Extra calls, lost selling time, lower customer trust, discounts or cancelled orders.

A range with reasons helps

If the data are current and the reasons for the date can be seen.

EVIDENCE OF RELEVANCE

Described by participants, or still an assumption

Described in the interviews

A date should be confirmed with production before it is given. **FR-04**

The manager needs the date, order status, queue, load by station and material availability. **FR-05, FR-08**

Trust depends on current data and visible reasons for a change. **FR-05**

Delays come from materials, load, breakdowns and changed orders. **FR-09**

Not measured yet: assumptions

Share of orders delivered late. No company statistics are available.

Duration of each stage. Taken from the literature, documented and configurable (NFR-07).

Supplier lead times. Modelled as a controlled share of late deliveries.

Tools in use today. To be confirmed at the company.

LITERATURE REVIEW

20 articles, four themes, a common limitation

20
peer-reviewed articles, 2021 to 2025

10 + 10
original studies and reviews

3
databases: Scopus, Web of Science, Google Scholar

Theme	Typical method	Main limitation
Delivery date	Review, MILP model	Not furniture-specific, no showroom input
Configuration	Case study, concept	Conceptual or single-company evidence
Integration	System analysis	Starts after the order enters the factory
Twins and scheduling	Simulation, agents	Offline, or starts from known orders

RESEARCH GAP

No reviewed study connects the showroom order, a digital twin and a delivery commitment

Showroom to factory

Information transfer is fragmented. The proposed framework is conceptual and not tested.

Łukaszewicz et al.

Digital twin of a furniture line

Used for what-if analysis of load, not for each incoming order.

Lalik et al.

Order acceptance with capacity

Shown outside furniture and without a configurator order.

Cavero et al.

> Our project closes this gap and compares the accuracy of the digital-twin commitment with two simpler baselines.

EXISTING SYSTEM COMPARISON

Existing tools cover parts of the problem, none gives a date with reasons

System	Furniture focus	Order entry	Capacity view	Date estimate	Simulation	Reasons
S1 Klipboard (ERP)	Yes	Partly	Yes	Not stated	Not stated	Not stated
S2 imos (CAD/CAM, ERP)	Yes	Yes	Not stated	Not stated	Not stated	Not stated
S3 Allmoxy (ordering)	Yes	Yes	Partly	Not stated	Not stated	Not stated
S4 Tacton (configurator)	No	Yes	Not stated	Not stated	Not stated	Not stated
S5 Siemens Plant Simulation	No	Not stated	Yes	Partly	Yes	Not stated

5 systems, 8 criteria (6 shown here). Based on vendor documentation; “Not stated” means the source does not mention it, not that the feature is missing.

TECHNOLOGY COMPARISON

Four decisions, each justified by a requirement

Decision	Choice	Alternatives	Main reason
Backend	FastAPI	Django REST, Flask	Order schema is validated and documented automatically
Database	PostgreSQL	SQLite, MongoDB	JSON order plus relations, several writers at once
Simulation	SimPy	AnyLogic, Salabim	Plain Python, called from the backend, fixed seeds
Frontend	React, TypeScript, Vite	Vue 3, Jinja2 with HTMX	Typed contract with the API, separate container

Also used: Docker Compose (one-command start), pytest and GitHub Actions (tests on every pull request).

REQUIREMENTS SPECIFICATION

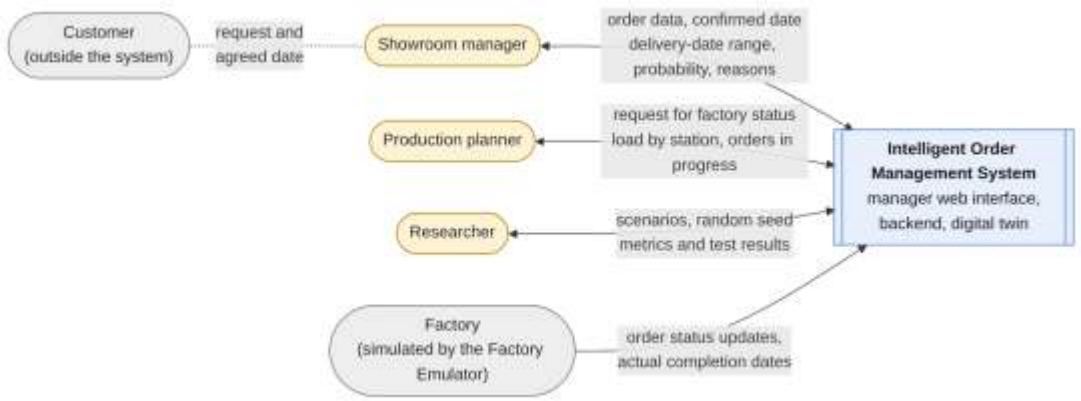
Every requirement can be verified

17 functional (13 must, 3 should, 1 could) · **9 non-functional** · each with an acceptance criterion

ID	Requirement	How it is verified
FR-04	Date range and probability of meeting the requested date	3 dates and a probability per estimate
FR-05	Reasons: bottleneck stations, queues, material delays	At least one reason per estimate
FR-11	Baselines B1 and B2 for the same orders	Metrics for twin, B1 and B2
NFR-01	Estimate within 10 s for 200 Monte Carlo runs	Timing test, 95% within 10 s
NFR-02	Order entered and confirmed in under 3 minutes	Walkthrough with 3 testers
NFR-06	Whole system starts with docker compose up	Clean-install test

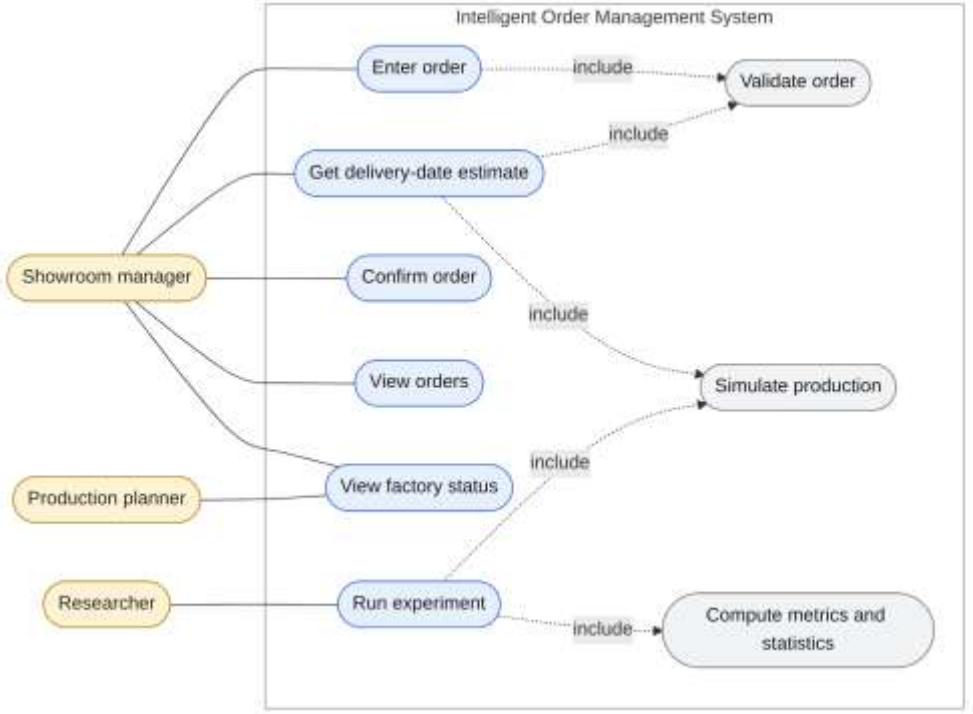
ARCHITECTURE AND DIAGRAMS

Who uses the system and what it does



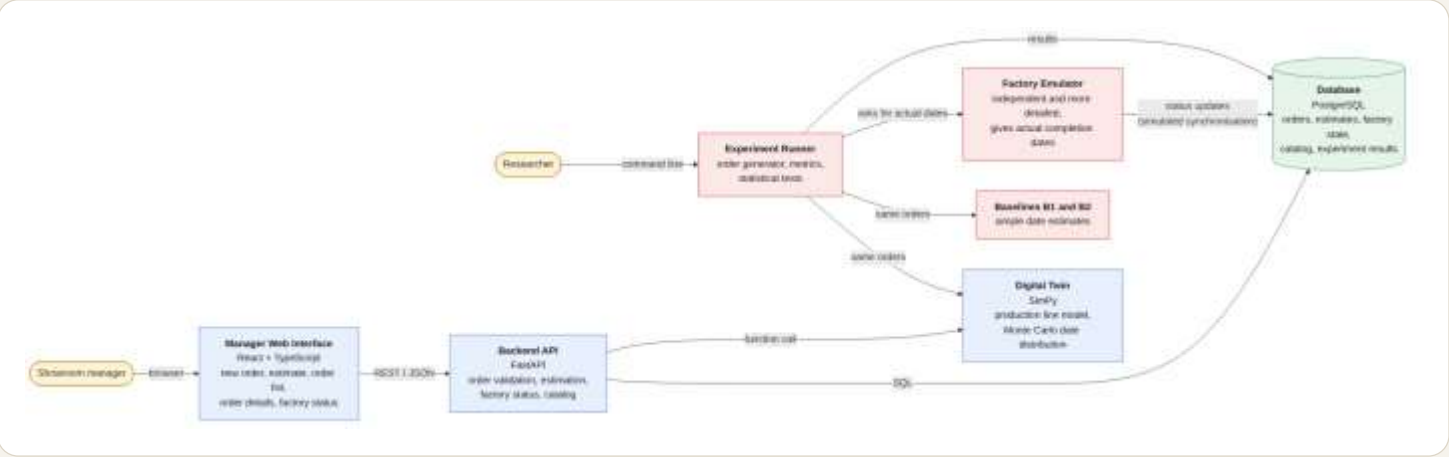
Actors: showroom manager, production planner, researcher.

Boundary: factory data come from the factory, simulated by the emulator.



ARCHITECTURE AND DIAGRAMS

Seven components, one idea: twin and emulator are separate



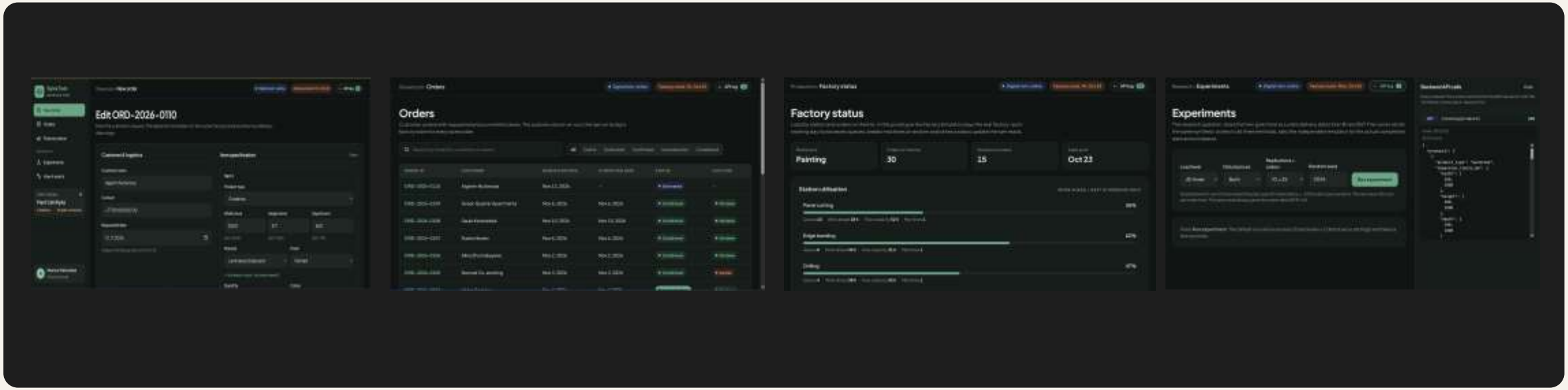
Twin is not emulator

The twin makes the estimate. A separate, more detailed emulator plays the real factory and gives the actual dates.

1 Manager enters the order → **2** Backend validates it → **3** Twin runs about 200 simulations → **4** Range, probability, reasons → **5** Manager confirms the date

UX AND PROTOTYPE

Four key screens and the API behind them



API EXAMPLE, ILLUSTRATIVE VALUES · POST /ORDERS/ESTIMATE

```
{"earliest": "2026-11-24", "most_likely": "2026-11-28", "latest": "2026-12-04", "probability_on_requested_date": 0.72, "assessment": "at_risk"}
```

VERIFICATION AND EVALUATION PLAN

How we will show the twin is better than the baselines

Three methods

Digital twin: 200 Monte Carlo runs, commits to the date that 80% of runs meet.

B1: average historical lead time plus a 20% buffer.

B2: static capacity check without factory feedback.

Five metrics

MAE of the committed date (H1)

On-time share of orders (H2)

MAPE and **bias** of the estimate

Estimation time, target 10 seconds or less

Conditions

6 scenarios: 3 load levels × 2 disturbance settings.

About 30 replications of 200 orders each, the same orders for all methods.

Actual dates come from an independent emulator.

Wilcoxon signed-rank test (one-sided, $\alpha = 0.05$) on paired replications, with Holm correction over 24 tests. A hypothesis counts as supported if it holds in at least 4 of 6 scenarios. Scenarios where the twin is not better are reported, not hidden.

PROJECT MANAGEMENT AND RISKS

Responsibilities, milestones and risks are written down

Responsibilities

Zhannur Meirambek: requirements, architecture, digital twin, frontend, statistics.

Ayazhan Orynbasar: backend, emulator, baselines, experiments, project plan.

Milestones

24 Oct 2026 Pre-defense 1

17 Dec Digital twin and emulator

24 Dec Backend and interface

7 Jan 2027 Experiments and statistics

14 Jan Article. Later submissions: dates from the department.

Main risks

No real factory data. Documented assumptions, several parameter values.

Twin-to-reality gap. Separate emulator, limits stated.

Too much work. Small scope, main chain first.

16 YouTrack tasks, one assignee each. Every change goes through a branch DIP-N and a pull request reviewed by the other student. Submissions are tagged in Git.

PRE-DEFENSE-1

Known limitations and what comes next

Known limitations

Synthetic data: results apply to the simulated factory.

The factory link is simulated by the emulator, not real machines.

Ten-second estimate target is not measured yet.

One product family, six stations, two interviews at one company.

Outstanding tasks

Backend and database with tests.

Digital twin and independent emulator.

Baselines, experiment runner, six scenarios and statistics.

Manager web interface connected to the API.

Repository: github.com/zhannur17/furniture-order-management-dt · tag **pre-defense-1**

YouTrack: https://furniture-dt.youtrack.cloud/projects/DIP